THE JOURNAL OF PHYSICAL CHEMISTRY C

C: Energy Conversion and Storage; Energy and Charge Transport

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J. Phys. Chem. C, **Just Accepted Manuscript** • DOI: 10.1021/acs.jpcc.8b09850 • Publication Date (Web): 21 Nov 2018

Hybrid Organic−Inorganic Functionalized Dodecaboranes and their Potential Role in Lithium and Magnesium Ion Batteries

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Abstract

Currently, development of rechargeable Mg ion batteries is an important and hot topic of research. For its development, the major challenge is to find suitable stable electrolyte anions, which possess solubility in low−polarity solvents. In this context, new organic and hybrid organic−inorganic functional derivatives of closo dodecaborane dianion viz., $B_{12}X_{12}^{2-}$ (X = −C≡CH, −C≡C−CN, and −C≡C−BO) are proposed here using density functional theory. The second excess electron in $B_{12}(C{\equiv}C{-}CN)_{12}^{2-}$ and $B_{12}(C{\equiv}C{-}BO)_{12}^{2-}$ is very strongly bound with the ΔE_2 value of 4.90 and 5.14 eV, respectively, in the gas phase which is almost six times higher than that of $B_{12}H_{12}^{2-}$ (0.81 eV). The various other factors responsible for the high stability of these predicted dianions have been explained in detail. We have explored the implications of these stable dianions as electrolytes in the Li and Mg ion batteries and the results are found to be highly promising. Among all the dianions considered here, the $B_{12}(C{\equiv}C{-}BO)_{12}^{2-}$ and $B_{12}(C{\equiv}C{-}CN)_{12}^{2-}$ are the most suitable choice as the electrolyte of the Li and Mg ion batteries, as the Li^+/Mg^{2+} salt of these two dianion requires very less energy to dissociate into corresponding cation and anion. In addition, the oxidation potential of $B_{12}(C{\equiv}C{-}BO)_{12}^{2-}$ and $B_{12}(C{\equiv}C{-}CN)_{12}^{2-}$ dianions vs Mg^{2+}/Mg is very high (12.91 V and 12.58 V, respectively). Present work reveals that it is possible to design desired multiply charged stable anions for appropriate applications through suitable organic−inorganic functionalization.

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1. INTRODUCTION

Boron hydride clusters (borane) have gained huge popularity among scientists because of their wide range of applications in the field of medicine.[1,2] Boron neutron capture therapy (BNCT) is one of the most widely known medical applications of boron hydrides.[3] In recent years various pristine and doped boron clusters are shown to display fascinating structures, bonding and properties.[4–8] Boron clusters also play an important role in catalysis[9–11], superacid chemistry[12,13] and nonlinear optical material.[14,15] Among the borane family, closo dodecaborane dianion i.e., $B_{12}H_{12}^{2-}$ has gained huge popularity because of its stability against the auto detachment of its excess electrons in the gas phase.[16] This is the smallest member of the borane family, which is stable in the gas phase[17], and its first and second excess electrons are bound with an energy of 4.50 and 0.81 eV, respectively. Whereas, the smaller boranes $B_nH_n^{2-}$ (n = 6−11), as well as most other multiply charged anions are unstable in the gas phase due to the electrostatic repulsion caused by the excess electrons.[18]

Among various applications, boron clusters are proposed to be used in designing highly stable and non−corrosive magnesium battery electrolyte.[19] The rechargeable Li and lead−acid batteries[20] are well developed as compared to the Mg batteries, although Mg battery has various advantages over Li battery viz., high theoretical capacity, low cost, and high abundance. The Mg battery has enormous potential to meet the future energy demands. An additional advantage of Mg over Li is that the Mg metal is not prone to the dendrite formation.[21–27] Recently Tutusaus *et. al.* and Tang *et. al.* have experimentally demonstrated that the $CB_{11}H_{12}^{-}$ shows excellent properties as halogen−free anionic component of the electrolytes for Mg, Li and Na ion batteries.[28–30] Soon after, Zhao *et. al.* have theoretically predicted an extremely stable $B_{12}(CN)_{12}^{2-}$ dianion by substituting the −H of $B_{12}H_{12}^{2-}$ with −CN ligand and studied its implications as an electrolyte in the lithium and magnesium ion batteries.[31] Later Jena and co−workers have proposed another very stable dianion, viz., $B_{12}(SCN)_{12}^{2-}$,[32] for Mg ion batteries. Recently in 2018, Moon *et. al.* have theoretically predicted even more stable $B_{12}(BO)_{12}^{2-}$ dianion in the gas phase.[33] In the $B_{12}(BO)_{12}^{2-}$, the binding energy of the second excess electron is 5.89 eV[33], which is 0.5 eV higher than that of the $B_{12}(CN)_{12}^{2-}$ dianion. Therefore, this dianion may also find applications in Mg ion batteries. Various other stable $B_{12}X_{12}^{2-}$ (X = F–At, OH) derivatives[34–37] are also investigated. In spite of various advantages, the main difficulty with the Mg ion batteries is that the Mg metal reacts with most of the polar solvents as well as the electrolyte anions. The highly inert low−polarity solvent like tetrahydrofuran (THF) is reported to be the most suitable solvent for the Mg ion batteries.[21,22,38] Therefore, development of suitable electrolytes that are soluble in low–polarity solvents is required for Mg ion batteries.

In the present work, our main focus is to predict new stable derivatives of $B_{12}H_{12}^{2-}$ dianion using the concept of ligand engineering. Till now, only highly stable inorganic derivatives of $B_{12}H_{12}^{2-}$ have been predicted; herein we propose hybrid organic−inorganic derivatives of $B_{12}H_{12}^{2-}$ dianion. The main advantage of using hybrid organic−inorganic derivatives is that the solubility of an electrolyte in a suitable less−polar solvent can be tuned through modifying the organic group. Thus, the presence of the organic−inorganic functional group in $B_{12}H_{12}^{2-}$ dianion is expected to increase the solubility of $B_{12}X_{12}^{2-}$ in low−polarity solvents. Therefore, we studied the organic (−C≡CH) and hybrid organic−inorganic (−C≡C−CN and −C≡C−BO) functional derivatives of $B_{12}H_{12}^{2-}$ by using the first principle based density functional theory (DFT). These hybrid organic−inorganic functional derivatives of $B_{12}H_{12}^{2-}$ are not reported till date. However, various organic derivative of $B_{12}H_{12}^{2-}$ viz., $B_{12}(OR)_{12}^{2-}$[39–41] and $B_{12}(CH_3)_{12}^{2-}$[42] have been synthesized earlier. Similar to $B_{12}(CH_3)_{12}^{2-}$, various other organic derivatives of boranes[43,44] and carboranes[45–49] are synthesized in the last few years. Very recently, Hahn and co–workers have shown that the derivatization can enhance the stability of the carba–closo–dodecaborate anion ($CB_{11}H_{11}^{-}$) for the high voltage battery electrolyte.[50]

2. COMPUTATIONAL DETAILS

All the calculations have been performed by using TURBOMOLE−6.6 software.[51] All the clusters have been optimized using B3LYP functional[52,53] and def−TZVPP basis set[54] within the framework of DFT. The method is represented as B3LYP/DEF throughout the paper. We have used the superfine m4 grid for the geometry optimization. In the recent past B3LYP method has been shown to work very well for calculating the molecular properties of boron clusters.[31,32] In addition to B3LYP method, geometry optimization of all the systems is also performed with PBE0[55,56] and TPSSH[57,58] method. It has been shown earlier that PBE0 calculated values of ΔE_1 and ΔE_2 agree very well with the corresponding CCSD(T) values for BO ligand[33]. The electronic stability of dianion against the auto detachment of its excess electrons is investigated by calculating the binding energy of its first and second excess electrons using B3LYP, PBE0 and TPSSH method. All the calculations have been performed with B3LYP/DEF method unless otherwise mentioned. For all the systems the harmonic vibrational frequency has been calculated and all the clusters possess only real frequency values. Therefore, the predicted structures for all the systems represent the true minima on the potential energy surface. Atomic charges are calculated using natural population analysis (NPA)[59] method. The atom−in−molecule (AIM) analysis[60,61] has been performed to understand the nature of bonding present in the studied systems using Multiwfn[62] software.

3. RESULTS and DISCUSSIONS

3.1. Geometrical and Structural Properties of $B_{12}X_{12}^{2-/1-/0}$ Systems

To begin with, we optimized all the $B_{12}X_{12}^{2-}$ (X = −C≡CH, −C≡C−CN, and −C≡C−BO) dianions using DFT with B3LYP functional and def−TZVPP basis set. For comparing our results with the earlier findings, we have optimized the previously reported $B_{12}X_{12}^{2-}$ dianions (X = −CN, −BO) as well as parent dianion i.e. $B_{12}H_{12}^{2-}$ using the same method, though B3LYP functional was only used for the earlier work.[31] All the chosen ligands possess higher electron affinity (EA) value than that of the H atom as shown in Table 1. Also, it is interesting to note that the EA of −C≡C−CN and −C≡C−BO is higher than that of either −C≡C−H or –BO or –CN. The $B_{12}(C≡CH)_{12}^{2-}$ dianion is optimized in highly symmetric icosahedral geometry (Figure 1). However, the optimized geometry of its neutral and negative charged systems possesses lower symmetry (S_6 and C_i, respectively) as shown in Figures S1 and S2. The acetylide substituted $B_{12}(C≡CH)_{12}^{2-}$ dianion is very intriguing as it can be used as a precursor for the synthesis of long chain derivatives, bulky organic derivatives as well as hybrid organic−inorganic derivatives of boranes. Later, we have replaced the −H of the −C≡CH group with the −CN and −BO inorganic ligands to form hybrid organic−inorganic ligand substituted dodecaborane dianions, viz., $B_{12}(C≡C-CN)_{12}^{2-}$ and $B_{12}(C≡C-BO)_{12}^{2-}$, respectively, as H atom of acetylide group is easily replaceable with the nucleophilic ligands. Both the $B_{12}(C≡C-CN)_{12}^{2-}$ and $B_{12}(C≡C-BO)_{12}^{2-}$ dianions also possess highly symmetric icosahedral geometry (Figure 1) as their minimum energy structure, whereas their neutral and singly negative anion possess C_i symmetry (Figures S1 and S2). All the $B_{12}X_{12}^{2-}$ systems are also optimized with PBE0/DEF and TPSSH/DEF method and found to possess icosahedral geometry as a minimum energy structure. Moreover, singly negative and neutral systems are also optimized with PBE0/DEF and TPSSH/DEF method and possess lower symmetry as we obtained using B3LYP/DEF method.

The calculated average bond distances of the optimized structures of neutral, monoanion and di–anions of $B_{12}X_{12}$ (X = −H, −BO, −CN, −C≡CH, −C≡C−CN, and −C≡C−BO) are reported in Table S1 using B3LYP/DEF method. The cage−ligand bond distances (B−X) are smallest in $B_{12}H_{12}^{2-}$ (1.20 Å). However, in $B_{12}(BO)_{12}^{2-}$ dianion the B−X bond distance is the maximum (1.66 Å). While, in the remaining $B_{12}X_{12}^{2-}$ dianions, the B−X bond distances are intermediate (1.53 Å). The B−B bond distance, that is the distance between the B atoms of the B_{12} cage is slightly smaller in $B_{12}H_{12}^{2-}$ (1.78 Å) as compared to that in the corresponding substituted systems. In the neutral ($B_{12}X_{12}$) and singly negative charged ($B_{12}X_{12}^{-}$) systems, the calculated optimized B−X (1.19 − 1.66 Å) and B−B (1.78 − 1.82 Å) bond distances are very close to that in the corresponding doubly negative charged system, showing a very small effect of charge state of $B_{12}X_{12}$ on the optimized bond distances. The diameter of the dianions, $B_{12}X_{12}^{2-}$, defined as the distance between the two farthest atoms, are found to be 5.79, 8.79, 9.15, 11.03, 13.94 and 14.26 Å for X = –H, −CN, −BO, −C≡CH, −C≡C−CN and −C≡C−BO systems, respectively as shown in Table S1. The optimized B−X (1.21 − 1.66 Å), B−B (1.77 − 1.80 Å) bond distances and diameter (5.79 − 14.27 Å) of $B_{12}X_{12}^{2-}$ systems calculated using PBE0/DEF and TPSSH/DEF method are found to be close in values with the corresponding bond distances calculated using B3LYP/DEF method as shown in Tables S2−S3. Similarly, in the neutral ($B_{12}X_{12}$) and singly negative charged ($B_{12}X_{12}^{-}$) systems, the calculated optimized B−X (1.18 − 1.66 Å) and B−B (1.78 − 1.82 Å) bond distances calculated using PBE0 and TPSSH methods (Tables S2−S3) are matching with the corresponding bond distances calculated using B3LYP/DEF method.

3.2. Binding Energy of First and Second Excess Electrons of $B_{12}X_{12}^{2-}$

The binding energy of the excess electrons of $B_{12}X_{12}^{2-}$ is highly important property to explain the stability of doubly charged anions in the gas phase. The binding energy of the first and second excess electrons of $B_{12}X_{12}^{2-}$, represented as ΔE_1 and ΔE_2, respectively, is calculated by using equations 1 and 2 and are reported in Table 2.

$$\Delta E_1 = E(B_{12}X_{12}) - E(B_{12}X_{12}^{-})\text{....................}(1)$$

$$\Delta E_2 = E(B_{12}X_{12}^{-}) - E(B_{12}X_{12}^{2-})\text{.................}(2)$$

In $B_{12}(C{\equiv}CH)_{12}^{2-}$, both the excess electrons are strongly bound with ΔE_1 and ΔE_2 values of 4.76 and 1.74 eV, respectively. The ΔE_2 of $B_{12}(C{\equiv}CH)_{12}^{2-}$ is almost double than that of the

$B_{12}H_{12}^{2-}$ (Table 2). Moreover, in the $B_{12}(C{\equiv}C{-}CN)_{12}^{2-}$ the binding energy of the second excess electron increases significantly to 4.90 eV, which is almost six times higher than that of the $B_{12}H_{12}^{2-}$ (0.81 eV). The binding energy of the first excess electron is also very high in $B_{12}(C{\equiv}C{-}CN)_{12}^{2-}$ (7.40 eV) as compared to that in $B_{12}H_{12}^{2-}$ (4.50 eV). The ΔE_1 and ΔE_2 of $B_{12}(C{\equiv}C{-}CN)_{12}^{2-}$ is very close with the corresponding values of $B_{12}(CN)_{12}^{2-}$ (8.49 and 5.18 eV, respectively), which indicates that even if the distance between the B_{12} cage and −CN moiety is larger in the $B_{12}(C{\equiv}C{-}CN)_{12}^{2-}$ dianion, the effect of −CN group remain almost the same on the stability of dianion. It also implies that various other organic−inorganic hybrid derivatives of $B_{12}H_{12}^{2-}$ can be designed with significantly higher excess electron binding energies. An increase in the binding energy of the excess electrons in the $B_{12}(C{\equiv}CH)_{12}^{2-}$ and the $B_{12}(C{\equiv}C{-}CN)_{12}^{2-}$ systems (as compared to that in $B_{12}H_{12}^{2-}$) can be attributed to the much larger electron affinities of the $-C{\equiv}CH$ and $-C{\equiv}C{-}CN$ groups as compared to that of H atom and also to the larger diameters.

Similarly, $B_{12}(C{\equiv}C{-}BO)_{12}^{2-}$ is found to be highly stable against the auto detachment of its excess electrons with a very high value of binding energy (ΔE_1 = 7.72 and ΔE_2 = 5.14, eV). In this system, the binding energy of the second excess electron is almost six times higher than that of the $B_{12}H_{12}^{2-}$ (Table 2). Moreover, the ΔE_2 value of the $B_{12}(C{\equiv}C{-}BO)_{12}^{2-}$ (5.14 eV) is relatively close to the ΔE_2 value of highly stable $B_{12}(BO)_{12}^{2-}$ (5.68 eV) dianion and very close to that of the $B_{12}(CN)_{12}^{2-}$ (5.18 eV), representing a very high stability of the dianion.

The ΔE_1 and ΔE_2 values of $B_{12}X_{12}^{2-}$ systems calculated using PBE0/DEF method are slightly higher (0.1−0.3 eV) than that from the corresponding values calculated using B3LYP method (Table 2). However, with TPSSH method the ΔE_1 and ΔE_2 values of $B_{12}X_{12}^{2-}$ are slightly lower (0.1 − 0.3 eV) than that from the B3LYP calculated values (Table 2). Thus the ΔE_1 and ΔE_2 values of $B_{12}X_{12}^{2-}$ systems calculated using B3LYP method is intermediate between the values calculated using PBE0 and TPSSH methods. In all the three methods the trend of ΔE_1 and ΔE_2 values remained the same for all the substituted $B_{12}X_{12}^{2-}$ systems. The ΔE_1 and ΔE_2 values of $B_{12}(C{\equiv}C{-}CN)_{12}^{2-}$ are calculated to be 7.63 (7.24) and 5.14 (4.14) eV and of $B_{12}(C{\equiv}C{-}CN)_{12}^{2-}$ are 7.99 (7.55) and 5.37 (4.98) eV using PBE0 (TPSSH) methods. High values of ΔE_1 and ΔE_2 represent the higher stability of these dianions in gas phase. Similar to the B3LYP method the ΔE_1 and ΔE_2 values of $B_{12}(C{\equiv}C{-}CN)_{12}^{2-}$ and $B_{12}(C{\equiv}C{-}BO)_{12}^{2-}$ calculated using PBE0 and TPSSH methods are slightly lower (1.1 − 1.4

eV and 0.3 − 0.5 eV, ΔE_1 and ΔE_2, respectively) than that of $B_{12}(CN)_{12}^{2-}$ and $B_{12}(BO)_{12}^{2-}$ dianions, respectively.

3.3. Analysis of the Stability of HOMO of $B_{12}X_{12}^{2-}$

The high stability of these dianions in the gas phase can be explained with the help of the stability of the highest occupied molecule orbital (HOMO) of $B_{12}X_{12}^{2-}$ dianions. From Table 2, we can see that more is the lowering of the energy of the HOMO of dianions, higher is the stability of the corresponding dianions, though the linear relationship is not strictly monotonic in nature. In these substituted $B_{12}X_{12}^{2-}$ dianions, the ligands (X = −C≡CH, −C≡C−CN, and −C≡C−BO) provide larger volume for the delocalization of excess electrons, thereby decrease the repulsion between the excess electrons present in HOMO as compared to that in the $B_{12}H_{12}^{2-}$, that can be visualized from Figure 2. Moreover, in the HOMO of $B_{12}X_{12}^{2-}$ dianions (X = −C≡C−CN, −C≡C−BO) electrons are more localized on the −C≡C− group while, in $B_{12}X_{12}^{2-}$ dianions (X = −CN, −BO), the electrons are more localized on the −CN and −BO group.

3.4. Electronegativity of $B_{12}X_{12}^{-}$ and $B_{12}X_{12}$ Systems

The electronegativity of neutral $B_{12}X_{12}$ and monoanion $B_{12}X_{12}^{-}$ is another important factor, which can explain the higher stability of these substituted $B_{12}X_{12}^{2-}$ dianions. The electronegativity values of $B_{12}X_{12}$ and $B_{12}X_{12}^{-}$ are calculated by taking the average of ionization potential and electron affinity values using Mulliken electronegativity approach as well as by taking the average of the negative of HOMO and LUMO energy of $B_{12}X_{12}$ or $B_{12}X_{12}^{-}$ systems. It can be seen from Table 3 that as the electronegativity of the $B_{12}X_{12}$ and $B_{12}X_{12}^{-}$ increases, their tendency to accept electrons to form $B_{12}X_{12}^{-}$ and $B_{12}X_{12}^{2-}$, respectively, also increases. Therefore, the $B_{12}(BO)_{12}^{-}$ monoanion possessing the highest electronegativity forms the most stable $B_{12}(BO)_{12}^{2-}$ dianion followed by $B_{12}(CN)_{12}^{-}$, $B_{12}(C≡C−BO)_{12}^{-}$, $B_{12}(C≡C−CN)_{12}^{-}$ and $B_{12}(C≡CH)_{12}^{-}$, respectively. It is to be noted that the combined electronegativity of the neutral and monoanion of $B_{12}X_{12}$ is responsible for the higher stability of the corresponding $B_{12}X_{12}^{2-}$ dianion. Therefore, by using the concept of electronegativity of the $B_{12}X_{12}$ and $B_{12}X_{12}^{-}$ systems we can easily explain higher stability of $B_{12}(BO)_{12}^{2-}$ dianion as compared to that of $B_{12}(CN)_{12}^{2-}$, which becomes difficult to explain in terms of the electron affinity of these two −CN (3.81 eV) and −BO (2.36 eV) ligands.

3.5. Natural Population Analysis

Now it is very interesting to see how the charge distribution affects the stability of these substituted dianion. The presence of small negative charge on the B_{12} cage of substituted $B_{12}X_{12}^{2-}$ calculated using natural population analysis (NPA) as compared to that in the B_{12} cage of $B_{12}H_{12}^{2-}$ can explain the higher stability of most of the systems except in case of $B_{12}(BO)_{12}^{2-}$ (Table S4). From the Table S4 one can see that the as the electronegativity of the ligand bonded with the B_{12} cage increases, the negative charge present on the B_{12} cage decreases, whereas, the overall negative charge of the ligand increases. Moreover, relatively higher charge delocalization has been observed in −C≡C−CN, −C≡C−BO substituted $B_{12}X_{12}^{2-}$ as compared to that in the −CN, −BO substituted $B_{12}X_{12}^{2-}$. Therefore, the ligands of $B_{12}X_{12}^{2-}$ (−C≡C−CN, −C≡C−BO) possess lesser charge density than that of the $B_{12}(CN)_{12}^{2-}$ and $B_{12}(BO)_{12}^{2-}$ dianion.

3.6. Analysis of topological properties of $B_{12}X_{12}^{2-}$

Further to analyze the nature of bonding between the boron cage and ligands as well as the bonding within the ligand atoms, we have performed atom in molecule (AIM) analysis using Multiwfn software. The various bond critical properties namely; electron density (ρ), Laplacian of electron density ($\nabla^2\rho$), Lagrangian kinetic energy G(r), Potential energy density V(r) and Energy density E_d(r) have been calculated at the bond critical points (BCP) and are provided in Table S5. The nature of bonding has been analyzed by using the Boggs criteria[61] of the covalent bonding. In all the cases the value of $\rho > 0.1$ and $\nabla^2\rho < 0$ at the BCP, existing between cage and ligands, indicating the existence of covalent bonding between the cage and ligand atom. Similarly, the bonding between the cage atoms (B−B bonding) also possesses covalent character.

3.7. Li^+ and Mg^{2+} Salts of $B_{12}X_{12}^{2-}$

Finally, the possibility of using the $B_{12}(C{\equiv}CH)_{12}^{2-}$, $B_{12}(C{\equiv}C{-}CN)_{12}^{2-}$ and $B_{12}(C{\equiv}C{-}BO)_{12}^{2-}$ dianions in Li and Mg ion batteries are explored here. For this purpose the Li^+ and Mg^{2+} salts of $B_{12}(C{\equiv}CH)_{12}^{2-}$, $B_{12}(C{\equiv}C{-}CN)_{12}^{2-}$ and $B_{12}(C{\equiv}C{-}BO)_{12}^{2-}$ are optimized with real frequency values. In the optimized structures (as shown in Figure 3 and Figure S3) of $Li@B_{12}X_{12}^{-}$ ($Mg@B_{12}X_{12}$), the Li^+ (Mg^{2+}) ion is present at one of the triangular face of the $B_{12}X_{12}^{2-}$. In the $Li_2@B_{12}X_{12}$, two Li ions are present exactly at the opposite triangular faces of the $B_{12}X_{12}^{2-}$. The dissociation energy required to remove first Li^+ (ΔE_{Li1}) and second Li^+

(ΔE_{Li2}) from $Li_2@B_{12}X_{12}$ salt and Mg^{2+} ion (ΔE_{Mg}) from $Mg@B_{12}X_{12}$ salt is calculated with B3LYP and PBE0 methods by using equations 3, 4 and 5 and the values are provided in Table 4 and Table S7, respectively.

$$\Delta E_{Li1} = [E(Li@B_{12}X_{12}^{-})] + E(Li^{+})] - [E(Li_2@B_{12}X_{12})]............(3)$$

$$\Delta E_{Li2} = [E(B_{12}X_{12}^{2-}) + E(Li^{+})] - [E(Li@B_{12}X_{12}^{-})]...................(4)$$

$$\Delta E_{Mg} = [E(B_{12}X_{12}^{2-}) + E(Mg^{2+})] - [E(Mg@B_{12}X_{12})]................(5)$$

The dissociation energy of the Li^{+} and Mg^{2+} salt of the $B_{12}(C{\equiv}CH)_{12}^{2-}$ is quite higher (~ 1−3 eV) than that of the $B_{12}(CN)_{12}^{2-}$ and $B_{12}(BO)_{12}^{2-}$. On the other hand, the dissociation energy of the Li^{+} salt of the $B_{12}(C{\equiv}C{-}CN)_{12}^{2-}$ and $B_{12}(C{\equiv}C{-}BO)_{12}^{2-}$ is quite lower (~1.5 eV) than that of the $B_{12}(CN)_{12}^{2-}$ and $B_{12}(BO)_{12}^{2-}$ (Table 4). Similarly, the dissociation energy of Mg^{2+} salt of $B_{12}(C{\equiv}C{-}CN)_{12}^{2-}$ and $B_{12}(C{\equiv}C{-}BO)_{12}^{2-}$ is significantly smaller (~ 3 eV) than that of the $B_{12}(CN)_{12}^{2-}$ and $B_{12}(BO)_{12}^{2-}$, implying a better suitability of hybrid organic−inorganic group substituted $B_{12}X_{12}^{2-}$ dianion as an electrolyte for Li and Mg battery. It is noteworthy to mention that the presence of small charge density on −C≡C−CN, −C≡C−BO ligands due to their larger size and lesser electronegativity as compared to that on the −CN and −BO ligands is responsible for very weak interaction between the $B_{12}X_{12}^{2-}$ (X = −C≡C−CN, −C≡C−BO) and Li^{+}/Mg^{2+} ions. Moreover, in these salts, the metal ion is coordinated with the C atom of the ligand (Figure 3 and Figure S3), therefore, forming a weaker and less polar bond. Whereas, the high charge density on the −CN and −BO ligands as well as the bonding of the Li^{+}/Mg^{2+} ion with highly electronegative −N and −O atoms of these ligands (Figure S4) leads to the formation of more polar bond which may decrease the solubility of these salts in low−polarity solvents, most appropriate for Mg battery. Thus, a smaller dissociation energy of the Li^{+}/Mg^{2+} salts of $B_{12}X_{12}^{2-}$ (X = −C≡C−CN, −C≡C−BO), a smaller charge density on −C≡C−CN, −C≡C−BO ligands, Li/Mg−C coordination (vs Li/Mg−O/N coordination) and expected higher solubility of $B_{12}X_{12}^{2-}$ (X = −C≡C−CN, −C≡C−BO) in low−polarity solvent because of the presence of organic moiety, can make these dianions highly suitable for use in Mg^{2+} ion batteries.

Furthermore, in the THF solvent (epsilon = 7.52), the interaction of Li^{+} and Mg^{2+} ions with these dianions calculated using COSMO model also follow the same stability trend as we discussed above in gas phase. In THF solvent the dissociation energy of all the salts are significantly reduced as shown in Tables S6-S7. For Mg^{2+} salt of $B_{12}(C{\equiv}C{-}CN)_{12}^{2-}$, the solvent corrected dissociation energy is found to be negative (-0.17 eV) while for $B_{12}(C{\equiv}C{-}BO)_{12}^{2-}$ dianion the solvent corrected dissociation energy is almost zero, which represent the fully dissociative behaviour of both the salts . The fully dissociative nature of these salts in the THF solvent represents their higher conductivity in the solution. Similarly using PBE0 method the solvent corrected dissociation energy of Mg^{2+} salt of $B_{12}(C{\equiv}C{-}CN)_{12}^{2-}$ is negative (-0.01 eV), while for $B_{12}(C{\equiv}C{-}BO)_{12}^{2-}$ dianions the value is positive 0.14 eV. Thus again small negative and extremely small value of dissociation energy for Mg^{2+} salt of $B_{12}(C{\equiv}C{-}CN)_{12}^{2-}$ and $B_{12}(C{\equiv}C{-}BO)_{12}^{2-}$ dianions at PBE0/DEF level represents the fully dissociative behaviour of these salts. Moreover, the dissociation energy of Mg^{2+} salt of $B_{12}(CN)_{12}^{2-}$ and $B_{12}(BO)_{12}^{2-}$ in the solvent at both the PBE0/DEF and B3LYP/DEF levels are almost 2 eV higher than that of the Mg^{2+} salt of $B_{12}(C{\equiv}C{-}CN)_{12}^{2-}$ and $B_{12}(C{\equiv}C{-}BO)_{12}^{2-}$ dianions. All the calculated dissociation energy values clearly indicate that the solubility of the electrolytes derived from the inorganic-organic hybrid dianions, $B_{12}(C{\equiv}C{-}CN)_{12}^{2-}$ and $B_{12}(C{\equiv}C{-}BO)_{12}^{2-}$ is more suitable for Mg-ion batteries.

It is to be noted that for dianions the oxidation potential is calculated versus Mg^{2+}/Mg standard electrode potential of 2.05 V. While, for mono−anions the oxidation potential is reported versus Li^{+}/Li standard electrode potential of 1.37 V. As shown in Table 2, the oxidative potential of $B_{12}(C{\equiv}C{-}CN)_{12}^{2-}$ and $B_{12}(C{\equiv}C{-}BO)_{12}^{2-}$ dianions (12.58 V and 12.91 V, vs Mg^{2+}/Mg respectively) calculated in THF solvent is very high. Such a high oxidative potential of $B_{12}(C{\equiv}C{-}CN)_{12}^{2-}$ and $B_{12}(C{\equiv}C{-}BO)_{12}^{2-}$ dianion and their weaker interaction with the Li^{+} or Mg^{2+} ions, offers them as suitable candidates for the electrolyte of the high voltage Li and Mg ion batteries.

4. CONCLUSION

In summary, we proposed new hybrid organic−inorganic functional derivatives of the closo−dodecaborane dianions, viz., $B_{12}(C{\equiv}C{-}CN)_{12}^{2-}$ and $B_{12}(C{\equiv}C{-}BO)_{12}^{2-}$ using DFT. The high positive values of ΔE_1 and ΔE_2 shows the unique stability of these dianions in the gas phase. Moreover, both the $B_{12}(C{\equiv}C{-}CN)_{12}^{2-}$ and $B_{12}(C{\equiv}C{-}BO)_{12}^{2-}$ dianions possess very high oxidation potential and forms very weakly coordinated salt with Li^{+}/Mg^{2+} ion. Therefore, these dianions are proposed as better candidates as an electrolyte for the reversible Li and Mg ion batteries. Similar to the previously synthesized organic derivatives[42–50], proposed new dianions in this work can also be synthesized. The main advantage of using hybrid organic−inorganic derivatives is that the solubility of an electrolyte in a suitable less−polar solvent can be tuned through modifying the organic group. Moreover, the concept of hybrid organic−inorganic derivatives of dodecaborane proposed here not only opens up a new avenue to address the issue of electrolyte solubility in low–polarity medium (as required for Mg ion battery) through functionalization with various suitable organic groups but also provides a route in designing similar such appropriate hybrid materials for various other applications. We hope that the present findings will stimulate experimentalists to synthesize these highly stable dianions and explore their applications in the reversible Li and Mg ion batteries.

ASSOCIATED CONTENT

Supporting Information

Optimized bond distances of neutral, mono–anion and di–anion of $B_{12}X_{12}$ using B3LYP, PBE0, TPSSH methods; NPA charges of $B_{12}X_{12}^{2-}$ and AIM properties of $B_{12}X_{12}^{2-}$ at B3LYP/DEF level, dissociation energy of salts at PBE0/DEF and B3LYP/DEF levels. The optimized structures of neutral and mono–anion of $B_{12}X_{12}$ and optimized structures of Li^{+} and Mg^{2+} salt of $B_{12}X_{12}^{2-}$. This material is available free of charge via the Internet at http://pubs.acs.org.

ACKNOWLEDGMENTS

We would like to thank the Computer Division, Bhabha Atomic Research Centre for providing computational facilities. M.J. would like to thank Homi Bhabha National Institute for the Ph.D. fellowship in Chemical Sciences. It is a pleasure to thank Dr. P. D. Naik for his kind interest and continuous encouragement.

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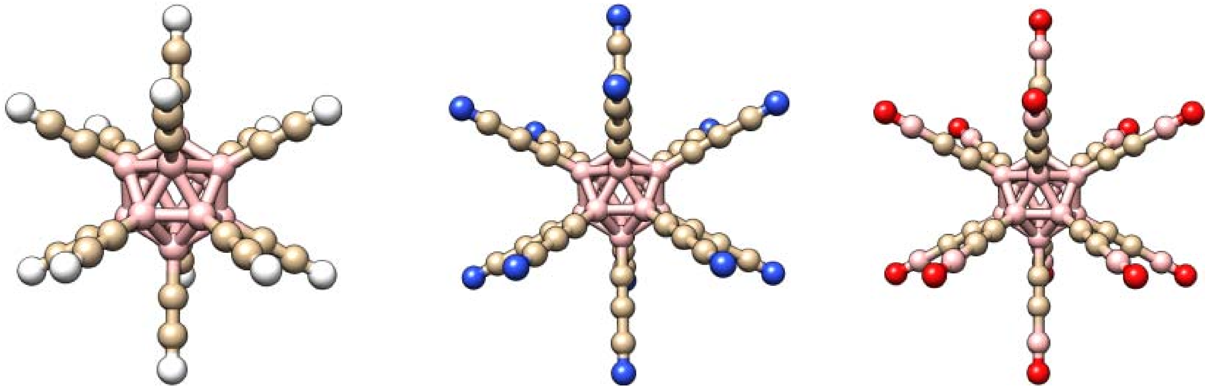
Figure Caption

Figure 1. Optimized structures of icosahedral $B_{12}X_{12}^{2-}$ (X = −C≡CH, −C≡C−CN, −C≡C−BO) dianions using B3LYP/DEF method. (Symmetry is provided within parenthesis)

Figure 2. The highest occupied molecular orbital (HOMO) pictures of $B_{12}X_{12}^{2-}$ dianions (X = −H, −BO, −CN, −C≡CH, −C≡C−CN, −C≡C−BO) obtained using B3LYP/DEF method

Figure 3. Optimized structures of $Li_2@B_{12}X_{12}$, $Li@B_{12}X_{12}^{-}$, $Mg@B_{12}X_{12}$ (X = −C≡C−CN) salts obtained using B3LYP/DEF method. (Symmetry is provided within parenthesis)



$B_{12}(C{\equiv}CH)_{12}^{2-}$ (I_h) $B_{12}(C{\equiv}C{-}CN)_{12}^{2-}$ (I_h) $B_{12}(C{\equiv}C{-}BO)_{12}^{2-}$ (I_h)

Figure 1. Optimized structures of icosahedral $B_{12}X_{12}^{2-}$ (X = $-C{\equiv}CH$, $-C{\equiv}C{-}CN$, $-C{\equiv}C{-}BO$) dianions using B3LYP/DEF method. (Symmetry is provided within parenthesis)

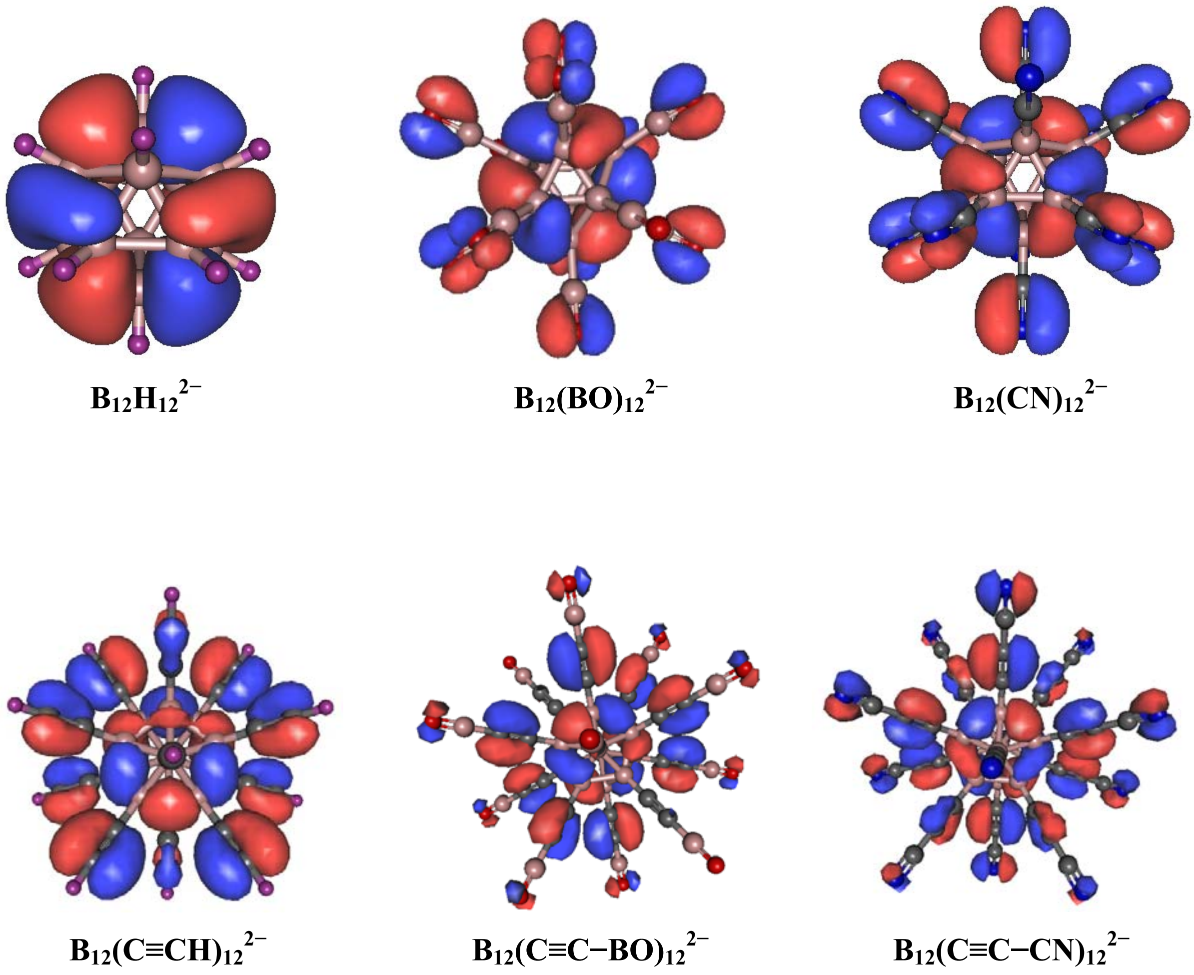
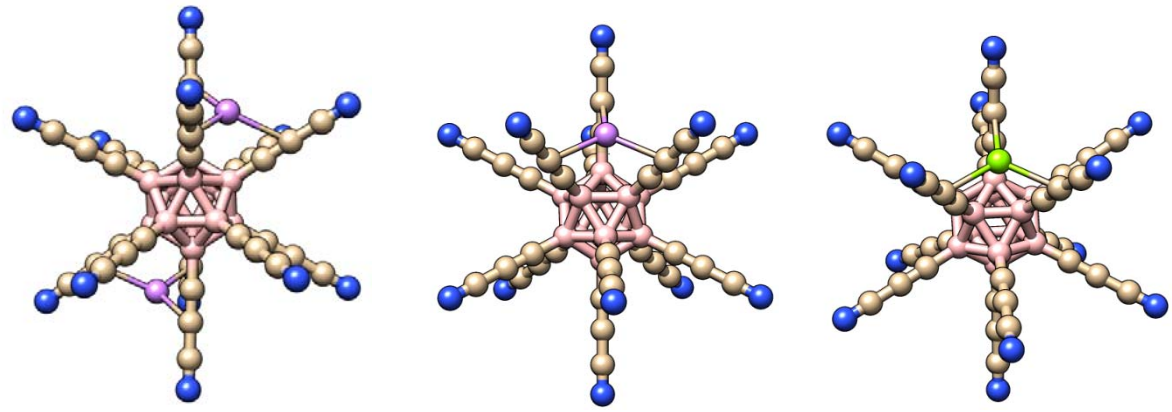


Figure 2. The highest occupied molecular orbital (HOMO) pictures of $B_{12}X_{12}^{2-}$ dianions (X = −H, −BO, −CN, −C≡CH, −C≡C−BO, −C≡C−CN) obtained using B3LYP/DEF method



$Li_2@B_{12}(C\equiv C-CN)_{12}$ (C_{3V}) **$Li@B_{12}(C\equiv C-CN)_{12}^{-}$ (C_1)** **$Mg@B_{12}(C\equiv C-CN)_{12}$ (C_1)**

Figure 3. **Optimized structures of $Li_2@B_{12}X_{12}$, $Li@B_{12}X_{12}^{-}$, $Mg@B_{12}X_{12}$ ($X = -C\equiv C-CN$) salts obtained using B3LYP/DEF method. (Symmetry is provided within parenthesis)**

Table 1. Calculated Values of the Vertical Electron Affinity (VEA, in eV) and Adiabatic Electron Affinity (AEA, in eV) of the Various Ligands X (X = −H, −CN, −BO, −C≡CH, −C≡C−CN, −C≡C−BO) using B3LYP/DEF Method. (Ligands are arranged in the Increasing Order of their Electron Affinity Value)

Ligands (X)	[a]VEA(AEA)
−H	0.12 (0.12)
−BO	2.36 (2.40)
−C≡CH	2.80 (2.87)
−CN	3.81 (3.81)
−C≡C−BO	4.11 (4.22)
−C≡C−CN	4.25 (4.33)

[a]VEA = E(X) − E(X^- at the optimized geometry of X)

[a]AEA = E(X) − E(X^-), both are at their optimized geometry

Table 2. Calculated Values of First (ΔE_1) and Second (ΔE_2) Electron Binding Energy (in eV), Energy of Highest Occupied Molecular Orbital (HOMO) (E_H, in eV) of $B_{12}X_{12}^{2-}$, Oxidation Potential[a] of Dianion $B_{12}X_{12}^{2-}$ ($E_{di(ox)}$[a], in V) and Monoanion $B_{12}X_{12}^{-}$ ($E_{m(ox)}$[a], in V) (X = −H, −CN, −BO, −C≡CH, −C≡C−CN, −C≡C−BO) using B3LYP/DEF Method. The ΔE_1 and ΔE_2 values calculated using PBE0 and TPSSH Methods are provided in Parenthesis and Square Bracket, Respectively.

Cluster	ΔE_1	ΔE_2	E_H	$E_{di(ox)}$[a] (Mg^{2+}/Mg)	$E_{m(ox)}$[a] (Li^+/Li)
$B_{12}H_{12}^{2-}$	4.50 (4.60) [4.42]	0.81 (0.96) [0.78]	0.58	9.15	4.51
$B_{12}(CN)_{12}^{2-}$	8.49 (8.76) [8.33]	5.18 (5.40) [5.00]	−4.03	15.62	8.14
$B_{12}(BO)_{12}^{2-}$	9.06 (9.38) [8.80]	5.68 (5.95) [5.44]	−4.52	16.12	8.29
$B_{12}(C\equiv CH)_{12}^{2-}$	4.76 (4.97) [4.66]	1.74 (1.96) [1.62]	−0.72	9.28	4.76
$B_{12}(C\equiv C-CN)_{12}^{2-}$	7.40 (7.63) [7.24]	4.90 (5.14) [4.74]	−4.04	12.58	6.32
$B_{12}(C\equiv C-BO)_{12}^{2-}$	7.72 (7.99) [7.55]	5.14 (5.37) [4.98]	−4.27	12.91	6.51

[a]Oxidation potential is calculated in THF solvent vs Mg^{2+}/Mg standard electrode potential of 2.05 V for the dianions and Li^+/Li of 1.37 V for the monoanions. A standard hydrogen electrode value of 4.42 V is used[63].($E_{di(ox)} = G_{sol}(B_{12}X_{12}) - G_{sol}(B_{12}X_{12}^{2-}) + IP_1 + IP_2 - 2.05$ vs Mg^{2+}/Mg and $E_{m(ox)} = G_{sol}(B_{12}X_{12}) - G_{sol}(B_{12}X_{12}^{-}) + IP_1 - 1.37$ vs Li^+/Li)[32]

Table 3. **Electronegativity Values[a] of Neutral $B_{12}X_{12}$ (χ_{neut}, in eV) and Monoanion $B_{12}X_{12}^{-}$ (χ_{anion}, in eV) (X = −H, −CN, −BO, −C≡CH, −C≡C−CN, −C≡C−BO) Calculated by Mulliken Electronegativity Scale[b] using B3LYP/DEF Method. The Electronegativity Values Calculated by taking the Average[c] of Negative of HOMO and LUMO Energy of Neutral $B_{12}X_{12}$ and Monoanion $B_{12}X_{12}^{-}$ are Provided within Parenthesis. (Systems are Arranged in the Increasing Order of their electronegativity Values)**

Neutral System	χ_{neut} [b,c]	ΔE_1 [d]	Monoanion System	χ_{anion} [b,c]	ΔE_2 [d]
$B_{12}@(C{\equiv}CH)_{12}$	6.30 (6.30)	4.76	$B_{12}@H_{12}^{-}$	3.10 (3.07)	0.81
$B_{12}@H_{12}$	6.53 (6.51)	4.50	$B_{12}@(C{\equiv}CH)_{12}^{-}$	3.25 (3.24)	1.74
$B_{12}@(C{\equiv}C{-}CN)_{12}$	8.62 (8.64)	7.40	$B_{12}@(C{\equiv}C{-}CN)_{12}^{-}$	6.14 (6.13)	4.90
$B_{12}@(C{\equiv}C{-}BO)_{12}$	9.00 (9.01)	7.72	$B_{12}@(C{\equiv}C{-}BO)_{12}^{-}$	6.42 (6.41)	5.14
$B_{12}@(CN)_{12}$	10.15 (10.16)	8.49	$B_{12}@(CN)_{12}^{-}$	6.82 (6.82)	5.18
$B_{12}@(BO)_{12}$	10.68 (10.72)	9.06	$B_{12}@(BO)_{12}^{-}$	7.34 (7.34)	5.68

[a]Electronegativity values are calculated by doing the single point energy calculations using the optimized geometry at the same level.

[b]χ=(IP+EA)/2 and [c]$\chi=[(-E_{HOMO}) + (-E_{LUMO})]/2$

[d]ΔE_1 and ΔE_2 values are given here to show the correlation between the electronegativity and binding energy of electron.

Table 4. The Dissociation Energy (in, eV) Required to Dissociate Metal Ion from its $M_n@B_{12}X_{12}$ Salt [M = Li^+ (n = 2), Mg^{2+} (n = 1); X = −C≡CH, −C≡C−CN, −C≡C−BO, −CN, −BO] Calculated using B3LYP/DEF Method

Cluster	ΔE_{Li1}	ΔE_{Li2}	ΔE_{Mg}
$M_n@B_{12}(C{\equiv}CH)_{12}$	6.18	8.57	21.01
$M_n@B_{12}(C{\equiv}C{-}CN)_{12}$	3.33	5.34	14.82
$M_n@B_{12}(C{\equiv}C{-}BO)_{12}$	3.31	5.35	14.86
$M_n@B_{12}(CN)_{12}$	4.72	6.91	17.47
$M_n@B_{12}(BO)_{12}$	5.11	7.20	17.99

TOC GRAPHICS